THESIS REVISION GUIDE
Sections marked 01.11

Concrete remaining tasks

This checklist covers the three sections marked `01.11: Polish!` in Chapter 3. Line numbers refer to `5_bandwidth_selection.tex`. Sections marked 02.11 and citation cleanup are deferred. Items focus on mathematical accuracy and useful shortening.

3.1 KDE preliminaries

☐ **Complete the consistency sentence inside the marked block.** Add the missing $\to 0$ in $\|K_{h_n} * f - f\|_2 \to 0$ and describe $h_n \to 0$ and $nh_n \to \infty$ as sufficient conditions for the bias and variance to vanish. [source lines 65] See Hansen, Section 2.2, Theorem 2.1.

3.3.1 Divergence matrices and the Perron vector

☐ **Repair the acute-chain lemma proof.** The diagonal entries of D are zero, so use positivity off the diagonal and irreducibility. The acute-chain assumption gives $\langle P_i, Q_i \rangle \geq 0$, not necessarily > 0. The last displayed maximum also needs squared norms. Supply the additional argument needed for the stated *strict* exclusion of interior maximizers, or weaken the conclusion to what the proof establishes. [source lines 283–344]

☐ **Shorten the same proof.** Replace the long expansion of μ by the one-line identity $\mu = x_1 + \sum_k a_k(x_{k+1} - x_k)$, then retain the essential comparison. Once the proof is complete, remove or fill the empty appendix subsection for the same lemma. [source lines 305–317; appendix lines 225–229]

☐ **Fix the Gaussian acute-chain calculation.** Explicitly use continuity of P_n and $P_n(0) = 1$. The Fourier-domain integrand in the positivity sentence must contain $b_k b_l$, as in the displayed equation, rather than $\delta_k \delta_l$. State the positive, strictly ordered bandwidth assumptions in the proposition. [source lines 351–369]

3.3.2 Similarity transformations

☐ **Repair the theorem's boundary assumption.** For completely monotone ϕ, require a finite continuous extension to 0 when applying it to $D(x, x) = 0$. In the Laplace representation include a possible atom at $t = 0$; this also covers constant functions. Nonconstant ϕ and distinct embedded points give the stated strict result. [source lines 383–413] See Guella, Theorem 3.1 and discussion following it.

☐ **Shorten the Schoenberg discussion.** Once the assumptions are correct, cite the existing theorem for the general equivalence and keep only the derivation or examples needed for the thesis. The Gaussian-kernel case is the one used in the experiments. [source lines 389–430]

☐ **Tidy the transformation's scope.** State $\tau > 0$ in the entropy minimization and that $S \to I$ as $\tau \downarrow 0$ requires $D_{ij} > 0$ for $i \neq j$. Keep the distinction between positive definiteness and entrywise positivity when explaining Perron–Frobenius. [source lines 432–446]

After the substantive revisions, compile the thesis and check these sections and cross-references. The linked sources support the review; bibliography cleanup remains deferred.